

Matplotlib Lab Program — Q33 & Q34

Question 33: Scatter Plot with Empty Circles

Aim

To write a Python program to draw a scatter plot with empty (hollow) circles, taking a random distribution in X and Y and plotting them against each other.

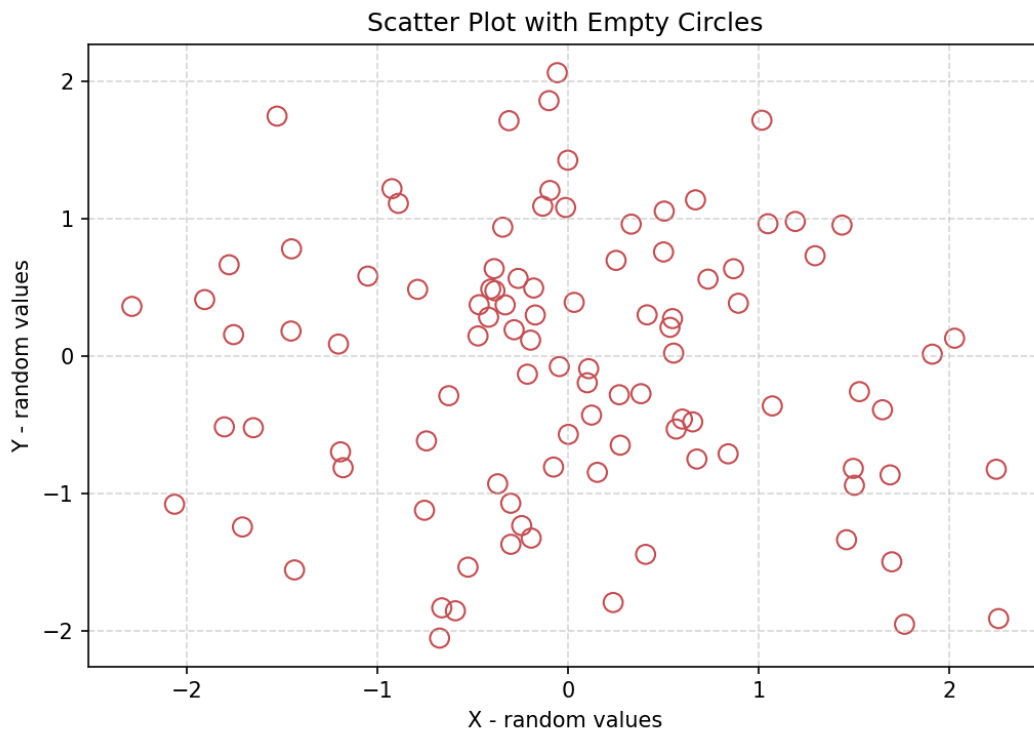
Source Code

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(7)
x = np.random.randn(100)
y = np.random.randn(100)

plt.figure(figsize=(7,5))
plt.scatter(x, y, s=80, facecolors='none', edgecolors='#C44E52')
plt.title('Scatter Plot with Empty Circles')
plt.xlabel('X - random values')
plt.ylabel('Y - random values')
plt.grid(True, linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig('imgs/q33_output.png', dpi=150)
print("Scatter plot with empty circles created and saved successfully.")
```

Output



Result

The Python program was executed successfully and a scatter plot with unfilled/empty circular markers was created for 100 randomly generated X and Y values.

Question 34: Scatter Plot with Balls of Different Sizes

Aim

To write a Python program to draw a scatter plot using random distributions to generate balls (markers) of different sizes and colors.

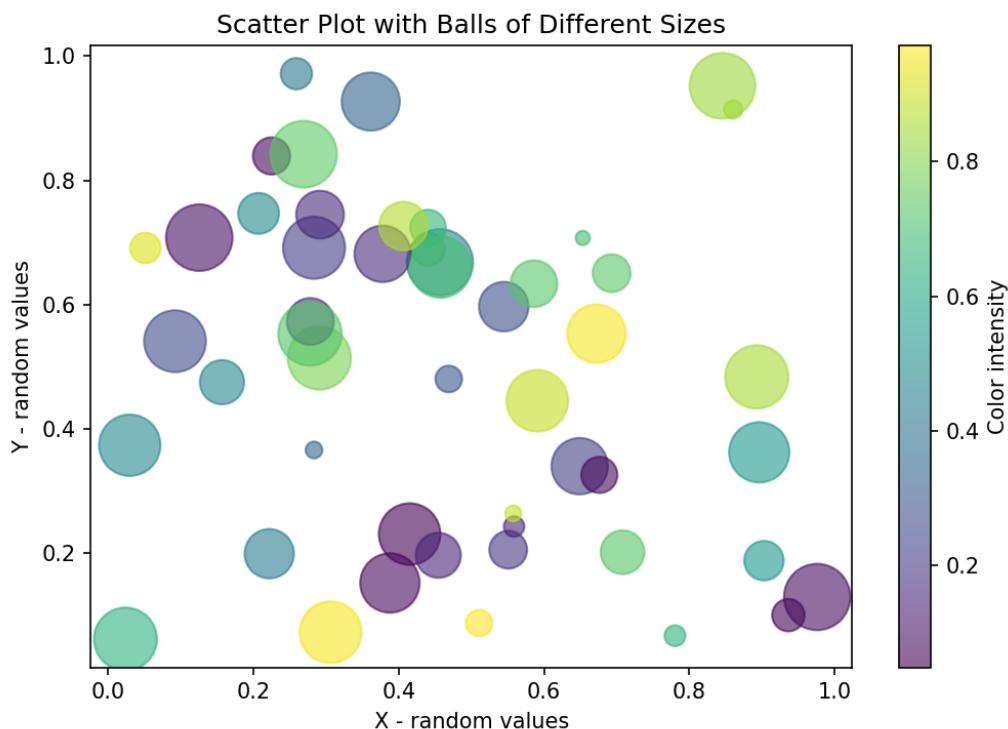
Source Code

```
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(3)
x = np.random.rand(50)
y = np.random.rand(50)
colors = np.random.rand(50)
sizes = 1000 * np.random.rand(50)

plt.figure(figsize=(7,5))
plt.scatter(x, y, c=colors, s=sizes, alpha=0.6, cmap='viridis')
plt.title('Scatter Plot with Balls of Different Sizes')
plt.xlabel('X - random values')
plt.ylabel('Y - random values')
plt.colorbar(label='Color intensity')
plt.tight_layout()
plt.savefig('imgs/q34_output.png', dpi=150)
print("Scatter plot with different sized balls created and saved successfully.")
```

Output



Result

The Python program was executed successfully and a scatter plot was produced in which each marker (ball) has a randomly generated size and color, visualized using a color map.